

USABILITY TESTING AND ANALYSIS

A Usability Evaluation of the Canara Engineering College Website

Submitted by
Ashwini Shenoy

1. Introduction

Usability testing is a user-centered evaluation method used to determine how easily people can use a digital product to complete specific tasks. It helps identify navigation problems, unclear labels, confusing layouts and other difficulties that users may experience.

This project evaluates the usability of the Canara Engineering College website. A small group of users is asked to complete common information-finding tasks while their success, completion time and difficulties are observed. The observations are then analyzed and converted into practical design improvement suggestions.

1.1 Aim

To evaluate the ease of use, navigation and overall usability of a college website by observing users while they complete common tasks.

1.2 Objectives

- Prepare a simple usability test plan and realistic user scenarios.
- Recruit a small sample of users and conduct usability testing sessions.
- Record task completion, approximate time and user difficulties.
- Identify usability issues from the observations.
- Suggest practical improvements to enhance the user experience.

1.3 Scope

The testing focuses on basic website navigation and information finding. It does not evaluate the technical implementation, server performance, security or backend code.

2. Usability Test Plan

Item	Details
Digital product	Canara Engineering College website
Testing method	Task-based usability testing
Target users	Students, friends or other people familiar with basic web browsing
Suggested sample size	5 users
Device	Laptop or smartphone
Session duration	Approximately 10–15 minutes per user
Data recorded	Task success, completion time, errors/difficulties and comments
Testing environment	Quiet environment without giving step-by-step help

2.1 Test Scenarios

Each participant is given the following tasks. The tester should avoid explaining where the information is located, because the purpose is to observe how naturally the user can find it.

Task	User Scenario	What to Observe
T1	Find the courses/programs offered by the college.	Navigation path and ease of finding academic information.
T2	Find information about admissions.	Whether the admission section is easy to locate.
T3	Find the college contact information.	Visibility of Contact/Contact Us information.
T4	Find the latest news, events or announcements.	Ease of locating current updates.

2.2 Instructions Given to Users

“Please use the website normally and complete each task. I will observe how you navigate the website. There are no right or wrong answers. Please think aloud if possible and tell me if anything is confusing.”

3. Participants and Data Collection

A small convenience sample of five users can be used for this academic usability study. Participants should represent typical people who may visit a college website. The tester should record observations during each session.

User ID	User Type	Device	Notes
U1	Student	Laptop/Phone	To be filled after testing
U2	Student	Laptop/Phone	To be filled after testing
U3	Student	Laptop/Phone	To be filled after testing
U4	Friend/Peer	Laptop/Phone	To be filled after testing
U5	Family/Other User	Laptop/Phone	To be filled after testing

3.1 Observation Sheet

User	Task	Success (Y/N)	Time	Difficulty / Observation
U1	T1 – Courses	Y	32 sec	Found courses easily; navigation was clear
U1	T2 – Admissions	Y	48 sec	Checked the menu before finding admissions
U1	T3 – Contact	Y	38 sec	Found contact information after some searching.
U1	T4 – News/Events	Y	35 sec	Found latest updates with little difficulty Needed a few extra clicks to locate course
U2	T1 – Courses	Y	41 sec	Needed a few extra clicks to locate courses.
U2	T2 – Admissions	Y	52 sec	Admission link was not immediately noticeable.
U2	T3 – Contact	N	1 min 10 sec	Could not locate contact information without help.
U2	T4 – News/Events	Y	44 sec	Found admissions through main navigation.
U3	T1 – Courses	Y	29 sec	Course information was easy to locate.
U3	T2 – Admissions	Y	45 sec	Found admissions through main navigation.
U3	T3 – Contact	Y	55 sec	Spent time searching for Contact Us.
U3	T4 – News/Events	N	1 min 45 sec	Had difficulty identifying recent events.
U4	T1 – Courses	Y	38 sec	Navigation was understandable but required scrolling Could

				not find admission information quickly.
U4	T2 – Admissions	N	1min 08 sec	Found contact details after checking the menu.
U4	T3 – Contact	Y	43 sec	Latest updates were reasonably easy to find.
U4	T4 – News/Events	Y	39 sec	Found course information without help.
U5	T1 – Courses	Y	34 sec	Hesitated while choosing the correct menu option.
U5	T2 – Admissions	Y	50 sec	Contact option was difficult to notice.
U5	T3 – Contact	N	1 min 15 sec	Contact option was difficult to notice.
U5	T4 – News/Events	Y	47 sec	Found news/events after exploring the homepage.

4. Usability Metrics and Analysis

The following simple metrics can be used to analyze the sessions.

4.1 Task Success Rate

Task Success Rate = (Number of users who successfully completed the task ÷ Total number of users) × 100

4.2 Task Completion Time

Record the approximate time taken by each participant to complete a task. A longer time may indicate that users needed more effort or navigation steps, although time should always be considered together with the user's comments and behavior.

4.3 User Difficulties

Record repeated problems such as difficulty locating a menu item, unclear labels, excessive navigation, small text, poor visibility of important links or confusion between similar options.

4.4 Results Summary Template

Task	Successful Users	Success Rate	Average Time	Main Observation
T1 – Courses				
T2 – Admissions				
T3 – Contact				
T4 – News/Events				

Example calculation: If 4 out of 5 users complete a task successfully, the task success rate is 80%. Replace the values with the results from your actual testing.

5. Findings

After completing the testing, the observations should be grouped into common usability issues. The following headings can be used in the final analysis.

Finding 1 – Navigation

State whether users could understand the main navigation and move between sections without assistance.

Finding 2 – Information Findability

State which information was easy or difficult to locate, such as courses, admissions, contact details or events.

Finding 3 – Labels and Menus

State whether menu names and buttons were clear to users.

Finding 4 – Mobile/Laptop Experience

State whether users experienced any difficulty because of screen size, layout or scrolling.

Finding 5 – Overall User Experience

Summarize repeated comments, hesitation, backtracking or requests for help observed during the sessions.

5.1 Suggested Findings Format

Issue Observed	Evidence from Users	Impact on Usability
Example: Contact information was not immediately visible.	Several users took longer or searched through multiple sections.	Increases effort required to find basic information.
Example: Navigation labels caused confusion.	Users hesitated or went back to the previous page.	Makes information finding slower.
Add your actual issue	Add actual observation	Explain the effect on the user.

6. Design Improvement Suggestions

Recommendations should be directly connected to issues observed during testing. Possible recommendations for a college website include:

1. Make frequently needed links such as Admissions and Contact Us clearly visible.
2. Use simple and consistent navigation labels.
3. Group related information under clear menu categories.
4. Reduce unnecessary navigation steps for commonly requested information.
5. Ensure important content is easy to read on mobile screens.
6. Use clear headings, buttons and visual hierarchy to guide users.
7. Keep important announcements and events easy to find from the homepage.
8. Conduct another usability test after changes are implemented to verify whether the problems have improved.

6.1 Prioritization of Improvements

Improvement Area	Reason	Suggested Action
Navigation	Users may take longer to find information.	Simplify and organize the main menu.
Contact information	Basic contact details should be easy to locate.	Provide a clearly visible Contact Us option.
Admissions	Important for prospective students.	Place a direct Admissions link/button in a prominent location.
News and Events	Users need current updates.	Make recent announcements easy to access.
Mobile usability	Many users browse using smartphones.	Check layout, text size, buttons and scrolling on mobile.

7. Conclusion

This usability testing project provides a simple method for evaluating how users interact with a college website. By asking participants to complete realistic tasks and observing their behavior, the tester can identify problems that may not be obvious during normal website development.

The final results should be based on the actual testing sessions. Task success rates, completion times, user comments and repeated difficulties can be used to identify the most important usability issues. The resulting design suggestions can then be used to make navigation and information finding clearer and more efficient.